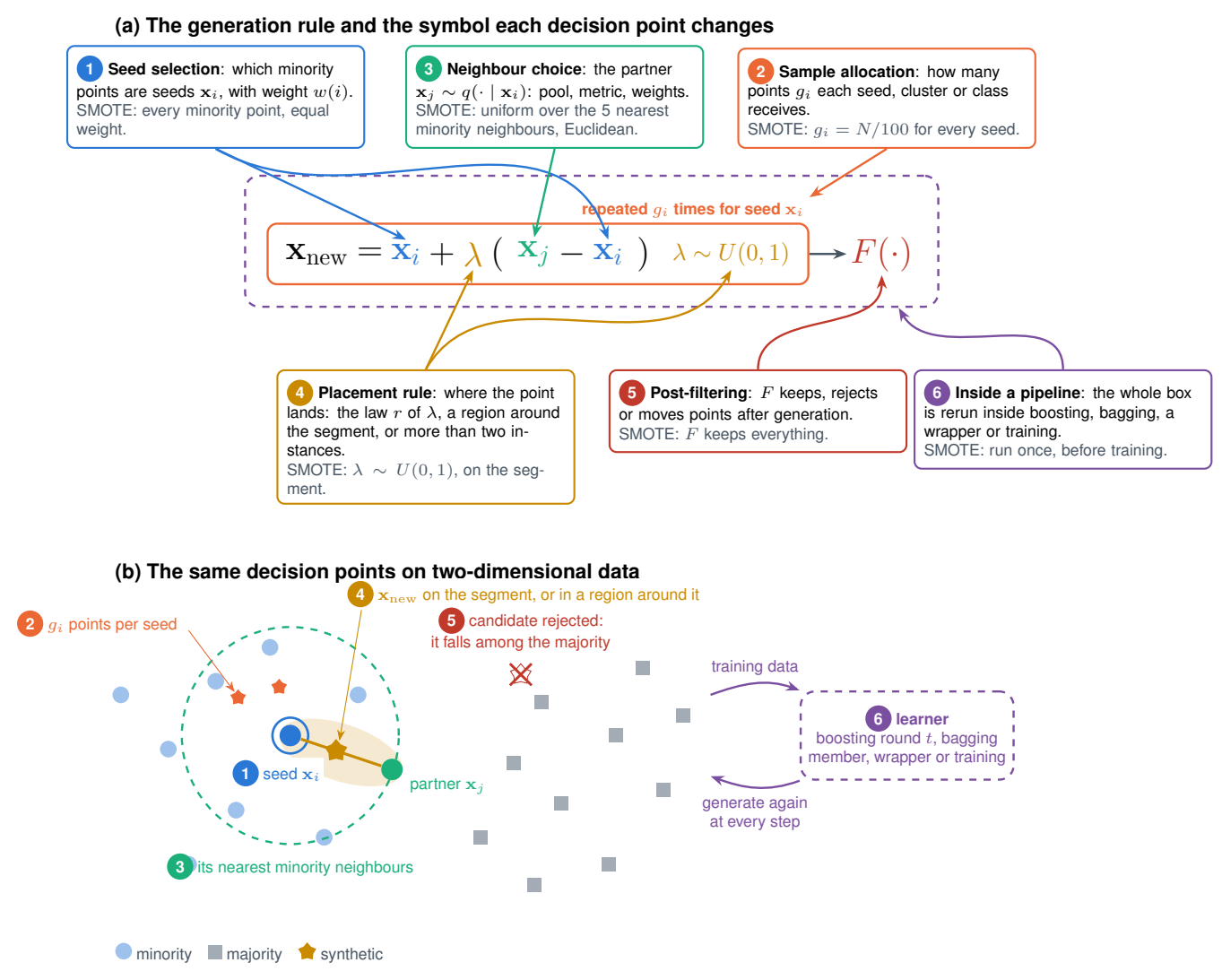

(a) The generation rule and the symbol each decision point changes
1 Seed selection: which minority points are seeds $\mathbf{x}_i$, with weight $w(i)$.
SMOTE: every minority point, equal weight.
3 Neighbour choice: the partner $\mathbf{x}_j \sim q(\cdot \mid \mathbf{x}_i)$: pool, metric, weights.
SMOTE: uniform over the 5 nearest minority neighbours, Euclidean.
2 Sample allocation: how many points g_i each seed, cluster or class receives.
SMOTE: $g_i = N/100$ for every seed.
repeated g_i times for seed $\mathbf{x}_i$
$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda \left(\mathbf{x}_j - \mathbf{x}_i \right) \quad \lambda \sim U(0,1)$
$F(\cdot)$
4 Placement rule: where the point lands: the law r of λ, a region around the segment, or more than two instances.
SMOTE: $\lambda \sim U(0,1)$, on the segment.
5 Post-filtering: F keeps, rejects or moves points after generation.
SMOTE: F keeps everything.
6 Inside a pipeline: the whole box is rerun inside boosting, bagging, a wrapper or training.
SMOTE: run once, before training.
(b) The same decision points on two-dimensional data
2 g_i points per seed
4 $\mathbf{x}_{\text{new}}$ on the segment, or in a region around it
5 candidate rejected: it falls among the majority
1 seed $\mathbf{x}_i$
partner $\mathbf{x}_j$
3 its nearest minority neighbours
training data
6 learner
boosting round t, bagging member, wrapper or training
generate again at every step
minority
majority
synthetic